



# **SECURE CHIME USING ARDUINO**

**ECB1223 – MICROCONTROLLERS AND INTERFACING**

**A PROJECT REPORT**

*Submitted by*

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*in partial fulfillment for the award of the degree*

*of*

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*in*

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**BONAFIDE CERTIFICATE**

Certified that this project report titled “**SECURE CHIME USING ARDUINO**” is the bonafide work of **ABITHA K R (2303811710622003), AKALYA B (2303811710622008), GAYATHRI K K (2303811710622026)** who carried out the project under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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## DECLARATION

We jointly declare that the project report on “**SECURE CHIME USING ARDUINO**” is the result of original work done by us and best of our knowledge, similar work has not been submitted to “**ANNA UNIVERSITY CHENNAI**” for the requirement of Degree of **BACHELOR OF ENGINEERING**. This project report is submitted on the partial fulfillment of the requirement of the award of Degree of **BACHELOR OF ENGINEERING**.

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# **CHAPTER 1**

## **PROBLEM STATEMENT**

### **1.1 INTRODUCTION**

The "Secure Chime Using Arduino" project is a smart and cost-effective security system designed to enhance safety at entry points such as homes, offices, schools, or restricted areas by combining modern sensor technology with microcontroller-based automation. Unlike traditional doorbells that simply notify someone's presence, this system provides an added layer of security by integrating motion detection, access verification, and real-time alerts. A PIR motion sensor detects movement near the entrance, triggering the Arduino to activate the chime system. To ensure that only authorized individuals gain access, the system can include modules such as RFID scanners, keypads, or fingerprint sensors for identity verification. Once the person is authenticated, a buzzer or chime is activated to alert the residents or personnel inside.

The system can be further enhanced with additional features like GSM modules for SMS alerts, LCD displays, or camera modules for image capture and monitoring. This project not only strengthens physical security but also offers a hands-on learning platform for students and hobbyists interested in embedded systems, IoT, and sensor-based automation, making it ideal for academic demonstrations, workshops, and practical applications in real-world security systems. The Secure Chime Using Arduino project also emphasizes energy efficiency and user customization, making it highly adaptable for different environments. The system can be programmed to operate only during certain hours, reducing unnecessary alerts and power consumption. It can also include features like sound level adjustment, automatic light activation upon detection, or even logging entry times into an SD card for record-keeping.

## 1.2 KEY ASPECTS OF THE PROJECT

The Secure Chime Using Arduino project incorporates several key aspects that make it a practical and educational solution for modern security needs. At the core of the system is motion detection using a PIR sensor, which identifies human presence near the entrance and activates the system. Once triggered, an access control mechanism—such as an RFID module, keypad, or fingerprint sensor—verifies the identity of the visitor, ensuring only authorized individuals are allowed access. The Arduino microcontroller plays a central role by processing sensor input and managing output devices like buzzers, LCD displays, or communication modules. A buzzer or speaker acts as the chime, notifying occupants of a visitor's presence, and this can be extended to include GSM modules for SMS alerts or IoT connectivity for remote access. The project may also include an LCD or OLED display to show messages like “Access Granted” or “Access Denied,” improving the user interface. An optional SD card module can be added to log entry data, recording time and identity for monitoring purposes. The system can be programmed to function only during specific hours to save energy and reduce unnecessary activations. Its modular design allows for easy customization, such as adding mobile app control, face recognition, or light automation. Overall, the project provides hands-on experience in embedded systems, Arduino coding, sensor interfacing, and real-world automation, making it ideal for students, hobbyists, and security enthusiasts alike. The system is also useful in hostels and PG accommodations to track movement at entry points and improve safety. In warehouses or storage facilities, it adds a layer of protection against unauthorized access, especially during off-hours. Furthermore, the project can be customized for use in medical clinics, server rooms, libraries, and small businesses where security and visitor tracking are necessary. Its affordability, scalability, and ease of customization make it suitable for both personal and institutional security applications. Schools, colleges, and labs can use the system to control access to staff rooms, exam halls, or restricted areas, ensuring that only authorized personnel or students can enter.

### **1.3 SMART ACCESS CONTROL AND ALERT SYSTEM USING ARDUINO**

The Smart Access Control and Alert System Using Arduino is an innovative and cost-effective solution designed to enhance security at entry points by integrating access verification and real-time alert mechanisms. This system combines the functionality of motion detection, identity verification, and alert notification into a single, compact unit controlled by an Arduino microcontroller. When a person approaches the door, a PIR motion sensor detects their presence and activates the system. The visitor is then prompted to authenticate using a method such as RFID scanning, keypad input, or fingerprint recognition. Upon successful authentication, access is granted and a buzzer or chime is triggered to alert the occupants. The system can also be configured to send SMS notifications via a GSM module, display messages on an LCD screen, or log entry data into an SD card for monitoring purposes. With the flexibility of Arduino, the project is easily customizable and scalable for homes, offices, schools, or other secure facilities. Its key advantages include low power consumption, affordability, easy programming, and potential integration with IoT-based platforms for remote access. This project not only strengthens physical security but also offers an excellent learning experience in automation, embedded systems, and smart technology applications. This project emphasizes modularity, energy efficiency, and ease of use, making it accessible to students, engineers, and hobbyists interested in embedded systems. It promotes the development of key skills in electronics, programming, and system design while addressing real-world challenges in physical security. As a result, the Smart Access Control and Alert System stands out as both a practical solution for securing entry points and an educational tool for understanding modern automation and security technologies.



## **CHAPTER 2**

### **DESIGN PROCEDURE**

#### **2.1 COMPONENTS USED**

##### **2.1.1 IR SENSOR (INFRARED SENSOR)**

The IR sensor is a key component used for motion or object detection. It consists of an infrared LED that emits IR rays and a photodiode or phototransistor that detects the reflected light. When a person or object passes in front of the sensor, the amount of reflected infrared light changes.

This change is captured and converted into an electrical signal which is sent to the Arduino. The IR sensor typically has three pins: VCC, GND, and OUT. The OUT pin goes HIGH (1) when motion is detected, signaling the Arduino to take action.

These sensors are very useful in automation and access control systems due to their low cost, ease of use, and reliability in detecting motion within a close range. In this project, it acts as the first line of sensing—alerting the system whenever someone approaches.



**Fig 2.1.1) IR Sensor**

### 2.1.2 ARDUINO UNO

The Arduino UNO is an open-source microcontroller board based on the ATmega328P chip. It contains 14 digital input/output pins, 6 analog inputs, a 16 MHz quartz crystal, USB connection, power jack, and reset button.

It is programmed using the Arduino IDE, which uses a simplified version of C/C++. In this project, the Arduino acts as the central control unit, receiving input from the IR sensor and giving output to the buzzer.

When the IR sensor detects motion, the Arduino processes this input and immediately sends a signal to activate the buzzer. The Arduino continuously runs a loop to monitor sensor status, ensuring real-time detection and response. It also allows for future upgrades like adding displays, relays, or wireless communication modules.



**Fig 2.1.2) Arduino UNO**

### 2.1.3 JUMPER WIRES

The Jumper wires are used to create electrical connections between components in a breadboard circuit. These wires carry power, ground, and signal between the Arduino and other devices like the IR sensor and buzzer.

They are available in various types and colors to make circuit assembly easier and more organized. In this project, jumper wires are crucial for connecting the IR sensor's output to a digital input pin on the Arduino, and connecting the buzzer to an output pin.

Without soldering, jumper wires provide a reusable and flexible means of creating and modifying circuits, especially during the testing phase.



**Fig 2.1.3) Jumper Wires**

#### **2.1.4 BUZZER**

The buzzer is an essential component in the cordless doorbell project, serving as the audio output device. It produces sound when activated, signaling the user that the doorbell button has been pressed.

Typically, a piezoelectric or electromagnetic buzzer is used for its compact size, low power consumption, and ease of integration into circuits. In this project, the buzzer is driven by the receiver circuit and provides a clear, audible chime, ensuring reliable notification for the user.



**Fig 2.1.4) Buzzer**

#### **2.1.5 USB CABLE AND LAPTOP WITH ARDUINO IDE**

The USB cable is used to connect the Arduino UNO to a laptop or desktop computer. It has two primary functions: uploading the program code and providing power to the board during development.

The Arduino IDE (Integrated Development Environment) is the software used to write, compile, and upload the code to the Arduino.

It supports multiple boards and libraries, making it suitable for both beginners and professionals.

In this project, the Arduino IDE is used to write the logic that reads the IR sensor's signal and activates the buzzer. It also allows real-time monitoring through the serial

monitor, which helps in debugging and testing.



**Fig 2.1.5) USB Cable**

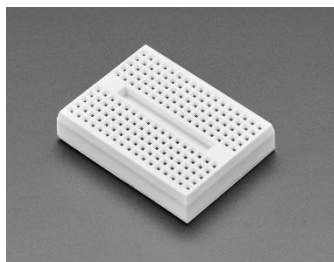
### **2.1.6 BREADBOARD**

The breadboard is a versatile and reusable platform used for prototyping and testing electronic circuits without soldering.

It consists of a grid of interconnected holes where components like resistors, LEDs, ICs, and wires can be easily inserted and rearranged.

In the cordless doorbell project, the breadboard facilitates quick assembly and testing of the transmitter and receiver circuits, allowing for modifications and troubleshooting.

Its flexibility and ease of use make it an essential tool for designing and refining electronic circuits.



**Fig 2.1.6) Breadboard**

## 2.2 ARDUINO PROGRAMMING LOGIC

The Arduino programming logic is the most crucial part of the Smart Access and Alert System. It defines how the microcontroller interprets sensor data and triggers an appropriate response. In this project, the goal is to detect any unauthorized or unexpected access using an IR sensor and alert the user via a buzzer. The IR sensor acts as the input, while the buzzer serves as the output. The logic is written using the Arduino IDE, which utilizes a simplified version of the C/C++ programming language.

The code is structured around two main functions: `setup()` and `loop()`. The `setup()` function runs only once when the Arduino is powered on or reset. In this function, pin modes are configured using the `pinMode()` command—typically setting the IR sensor's signal pin (e.g., pin 2) as INPUT and the buzzer's pin (e.g., pin 8) as OUTPUT. This sets the stage for real-time detection and response. The `loop()` function runs continuously and constantly checks the signal from the IR sensor using `digitalRead()`. When the IR sensor detects motion or the presence of an object, it outputs a HIGH signal. The Arduino receives this HIGH signal and immediately activates the buzzer by sending a HIGH signal to the output pin using `digitalWrite()`.

To make the alert more controlled and user-friendly, the code may include `delay()` functions, which limit the buzzer activation to a short duration such as 500 milliseconds. After the delay, the buzzer is turned OFF. This prevents continuous beeping and reduces noise. For enhanced system monitoring and debugging, the `Serial.begin(9600)` command is included in the setup to initialize serial communication. Using `Serial.print()` or `Serial.println()`, the Arduino can print real-time messages like "Motion Detected" or "No Motion" to the Serial Monitor. This feedback is especially useful during development and testing, helping to verify that the sensor is responding correctly and the buzzer is being triggered as intended.

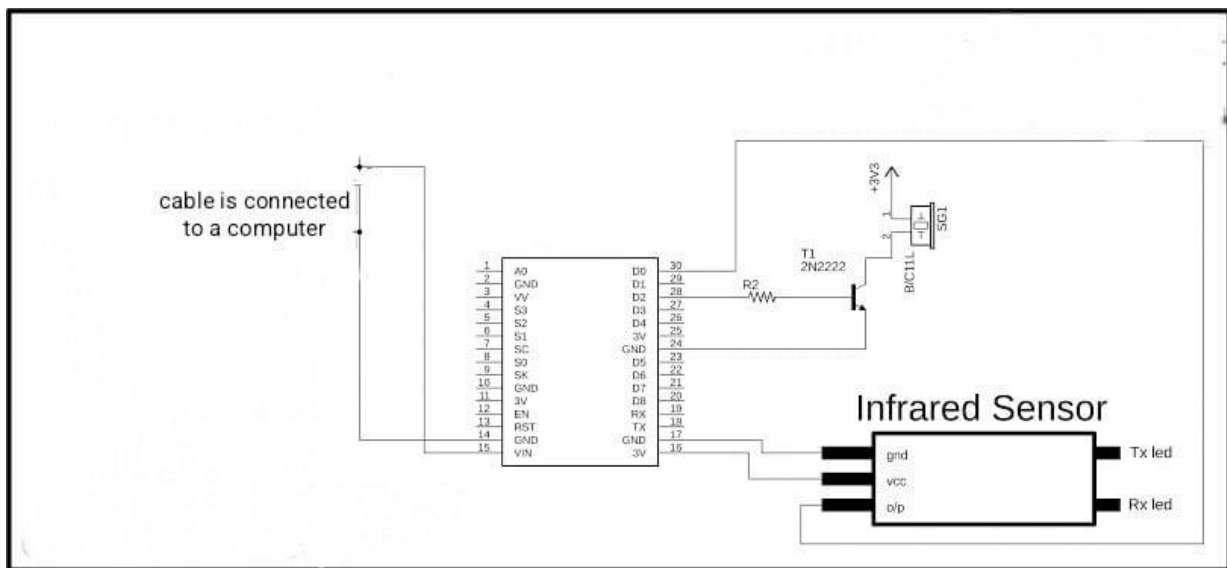
The logic is not only effective but also flexible. It can be expanded to include additional components or functions. For example, an LED can be added to provide a visual alert along with the sound, or a relay module can be integrated to control doors or lights based on sensor input. Moreover, the logic can be modified to send wireless alerts using Wi-Fi (ESP8266) or Bluetooth modules. Programmers can also add conditions to prevent false triggers, such as checking for repeated signals within a short time or using averaging methods to confirm genuine presence.

Uploading the code to the Arduino UNO is done via a USB cable connected to a laptop running the Arduino IDE. After writing and verifying the code, it is uploaded to the Arduino board using the "Upload" button. Once uploaded, the Arduino starts executing the logic automatically and can function independently without needing to stay connected to the laptop. This makes it a standalone system ideal for access control and security alert applications.

In summary, the Arduino programming logic is the heart of the system, enabling automated detection and response using clear and efficient code. The logic can be easily modified and expanded for additional features, making the project scalable and suitable for real-world applications like smart door systems, attendance tracking, and home security automation. Its simplicity also makes it a perfect entry-level project for students and hobbyists learning embedded systems and IoT.

## 2.3 CIRCUIT DIAGRAM

The circuit for the Smart Access and Alert System consists of an Arduino UNO connected to an IR sensor and a buzzer to detect motion and provide alerts. The IR sensor has three pins: VCC connected to the Arduino's 5V supply, GND connected to the common ground, and the output pin connected to one of the Arduino's digital input pins (e.g., pin 2) to send detection signals. The buzzer is connected to a digital output pin (e.g., pin 8) and ground, allowing the Arduino to activate it when motion is detected. All components share a common ground to ensure proper signal reference, and jumper wires and a breadboard are used to organize and connect the components without soldering. This simple yet effective circuit enables the Arduino to continuously monitor the IR sensor and sound the buzzer immediately upon detecting an object or person, providing a reliable alert system suitable for access control and security applications.



**Fig 2.3) Circuit Diagram of Secure Chime Using Arduino**



## **CHAPTER 3**

### **THE COMPONENTS**

#### **3.1 LIST OF THE COMPONENTS:**

1. Jumped wires (male to female)-5
2. Jumped wires (female to female)-5
3. Arduino UNO
4. Connecting cable
5. IR Sensor

#### **3.2 PRICE OF THE COMPONENTS:**

| <b>S.NO</b> | <b>NAME OF THE COMPONENTS</b>   | <b>PRICE</b> |
|-------------|---------------------------------|--------------|
| 1           | Jumped wires male to female pin | 100          |
| 2           | Jumped wires (female to female) | 100          |
| 3           | Arduino Uno                     | 349          |
| 4           | Connecting cable                | 289          |
| 5           | IR Sensor                       | 169          |

## **CHAPTER 4**

### **RESULT AND DISCUSSION**

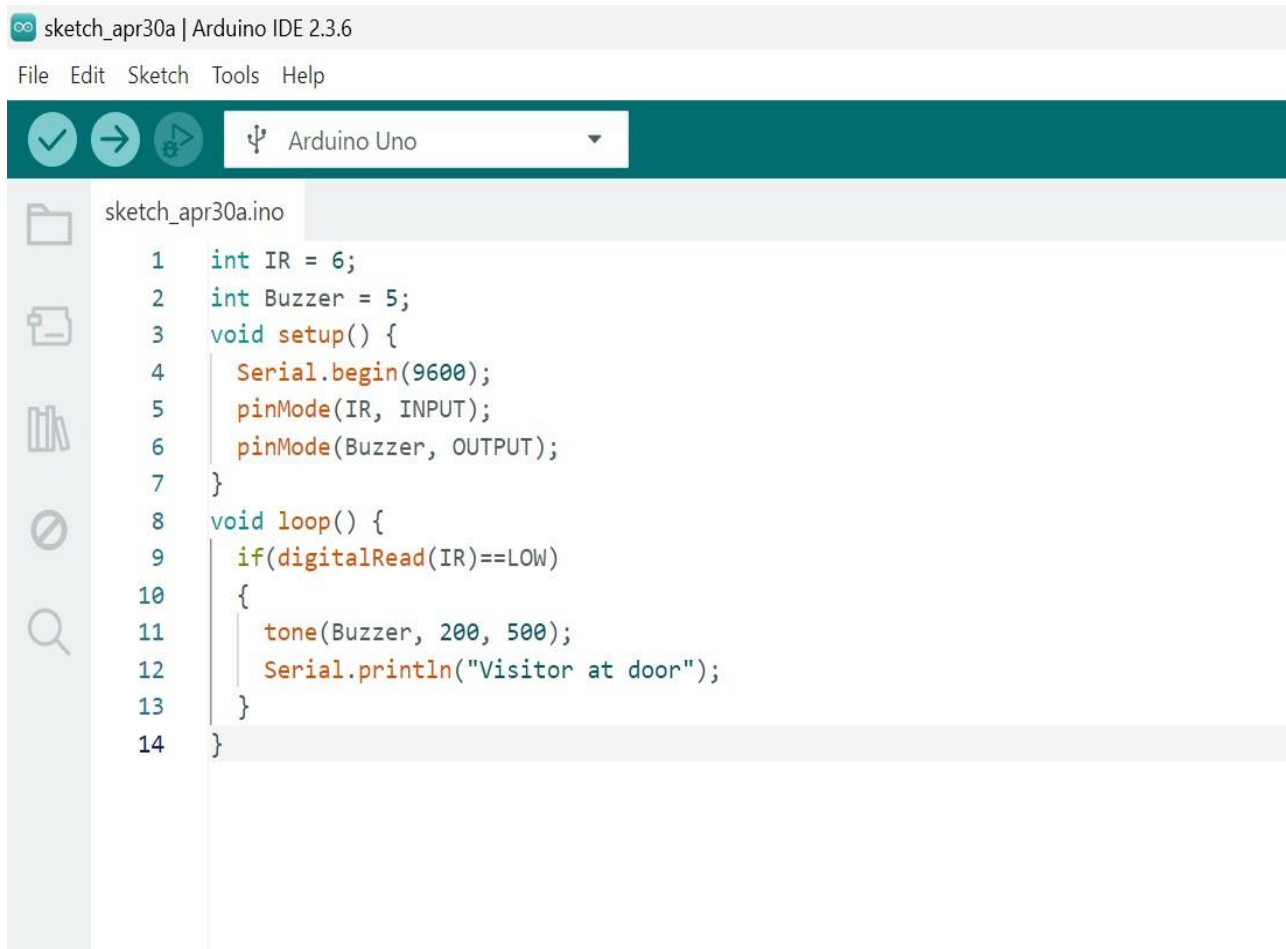
#### **4.1 WORKING MODEL**

The Secure Chime system using Arduino functions by detecting motion or the presence of an object using an infrared (IR) sensor module. The IR sensor emits infrared rays and monitors for reflections that occur when an object comes into its range. When someone or something enters the sensor's detection zone, it reflects the IR light back to the sensor, which then sends a digital HIGH signal to the Arduino board. This detection mechanism makes the system responsive to movement near a doorway or a restricted area.

Upon receiving the HIGH signal from the IR sensor, the Arduino processes the input and immediately activates a buzzer or chime connected to one of its output pins. The buzzer emits a sound, alerting the surroundings that motion has been detected. This feature is especially useful for alerting when someone approaches a door, enters a room, or accesses a restricted space. The use of a buzzer allows real-time audio feedback, ensuring that the alert is noticed without delay. The circuit remains simple and efficient, operating continuously to monitor the designated area.

To make the system more secure, it can be further enhanced by combining it with additional components such as RFID readers, keypads, or fingerprint sensors. This would allow the chime to only activate when both motion is detected and a valid authentication is provided, thereby preventing false alerts or unauthorized access. In its current form, however, the system serves as a reliable and contactless presence-based chime trigger using basic electronic components and Arduino logic, demonstrating a practical and low-cost security solution.

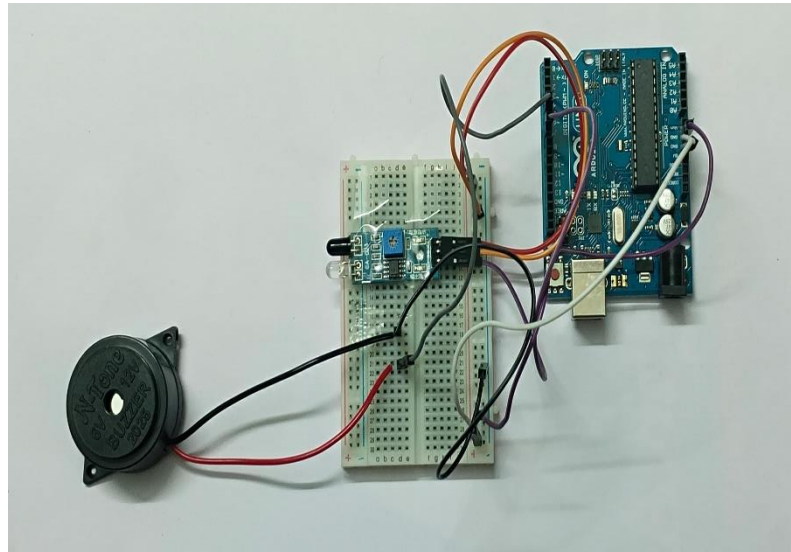
### 4.1.1 WORKING MODEL PROGRAM



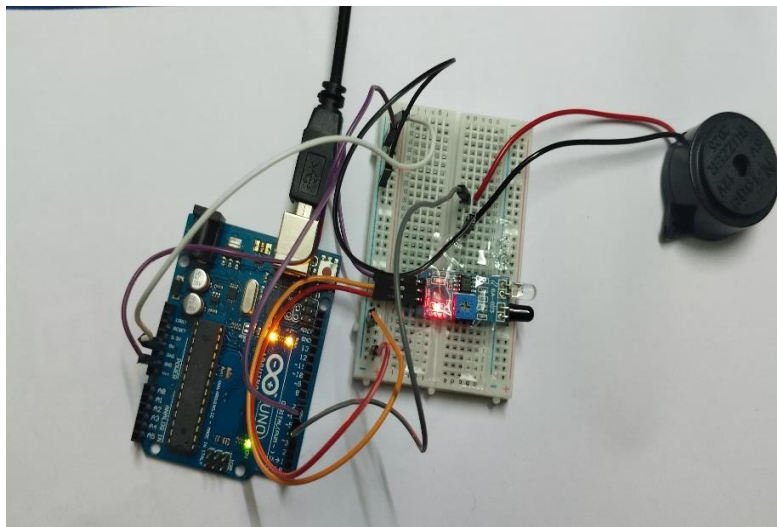
The screenshot displays the Arduino IDE 2.3.6 interface. The title bar shows 'sketch\_apr30a | Arduino IDE 2.3.6'. The menu bar includes 'File', 'Edit', 'Sketch', 'Tools', and 'Help'. The toolbar contains icons for checking, running, and uploading, along with a dropdown menu set to 'Arduino Uno'. The left sidebar shows a file explorer with a folder icon and a search icon. The main editor window displays the sketch 'sketch\_apr30a.ino' with the following C++ code:

```
1  int IR = 6;
2  int Buzzer = 5;
3  void setup() {
4      Serial.begin(9600);
5      pinMode(IR, INPUT);
6      pinMode(Buzzer, OUTPUT);
7  }
8  void loop() {
9      if(digitalRead(IR)==LOW)
10     {
11         tone(Buzzer, 200, 500);
12         Serial.println("Visitor at door");
13     }
14 }
```

**Fig 4.1.1) Working Model Program**



**Fig 4.1.2) Components Connection**



**Fig 4.1.3) Working Model**

## 4.2 CONCLUSION

The Secure Chime system using Arduino and an IR sensor offers a practical, low cost, and efficient solution for basic security and alerting needs. It operates by detecting motion or presence using the IR sensor and triggering a buzzer through the Arduino board, providing a real-time audible alert. This setup is particularly useful for monitoring entry points such as doors, gates, or restricted areas where human presence needs to be acknowledged without manual interaction. Its touchless nature enhances hygiene and ease of use, making it suitable for both personal and professional environments. This system not only demonstrates the effectiveness of combining sensors with microcontroller logic but also showcases how simple components can be used to build functional security solutions. It is ideal for beginners learning about Arduino, automation, and sensor integration. Furthermore, the design is highly flexible—additional components like RFID readers, keypads, GSM modules, or Wi-Fi connectivity can be integrated to provide authentication, remote alerts, and logging capabilities. In conclusion, the Secure Chime using Arduino is a strong example of how basic electronics and programming can be utilized to build a smart, responsive system. Its simplicity, adaptability, and potential for upgrades make it a valuable project in the field of embedded systems and security automation.